

Industrial Internship Report on

"Quiz Game"

Prepared by

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Executive Summary

This report presents the work carried out during my Industrial Internship provided by **Upskill Campus** and **The IoT Academy** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided practical exposure to Python Full Stack Development through the design and implementation of a real-world web application. We had to finish the project including the report in 6 weeks' time.

The project undertaken during this internship was **Quiz Game**, a web-based application developed using **Python Flask, SQLite, HTML, CSS, and JavaScript**. The application enables users to register, log in, participate in quizzes, receive automatic score evaluation, and view their results. An administrator can manage quizzes, questions, and user records through a dedicated admin interface.

Throughout the internship, I gained practical experience in project planning, frontend and backend development, database management, testing, debugging, and documentation. This internship enhanced my technical knowledge, problem-solving skills, and understanding of the complete software development life cycle.

Overall, the internship provided valuable industry exposure and strengthened my confidence in developing full-stack web applications.

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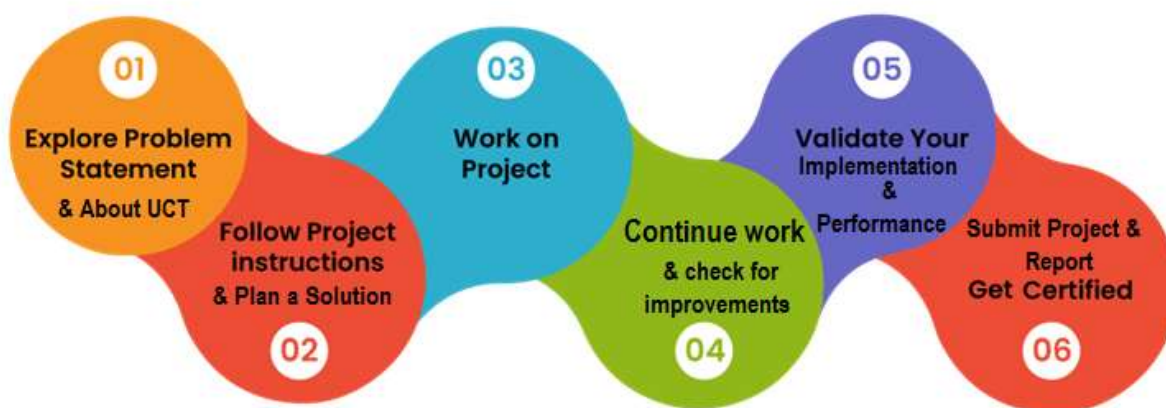
1 Preface

Industrial internships play a vital role in bridging the gap between academic learning and real-world software development. This internship provided an opportunity to understand industrial practices while applying programming concepts to solve practical problems.

During this internship, I developed a **Quiz Game** web application that allows users to register, log in, attempt quizzes, and instantly receive their scores. The system also provides an administrator with the ability to manage quiz questions, monitor user activities, and maintain quiz data efficiently.

The internship was planned in multiple phases, beginning with requirement analysis and project planning, followed by technology selection, application development, database integration, testing, debugging, and documentation. Each stage contributed to improving both technical and analytical skills.

How Program was planned



Working on this project enhanced my understanding of Python programming, Flask framework, SQLite database management, frontend technologies, authentication mechanisms, CRUD operations, debugging techniques, and software testing. It also improved my teamwork, communication, and project management abilities.

I sincerely thank **UniConverge Technologies Pvt. Ltd. (UCT)**, **Upskill Campus**, **The IoT Academy**, my internship mentors, faculty members, and everyone who guided and supported me throughout this internship. Their continuous encouragement and valuable feedback made this learning experience highly beneficial.

I encourage my juniors and peers to actively participate in industrial internships, as they provide practical exposure, improve technical competency, and prepare students for professional careers in software development.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT)**, **Cyber Security**, **Cloud computing (AWS, Azure)**, **Machine Learning**, **Communication Technologies (4G/5G/LoRaWAN)**, **Java Full Stack**, **Python**, **Front end** etc.



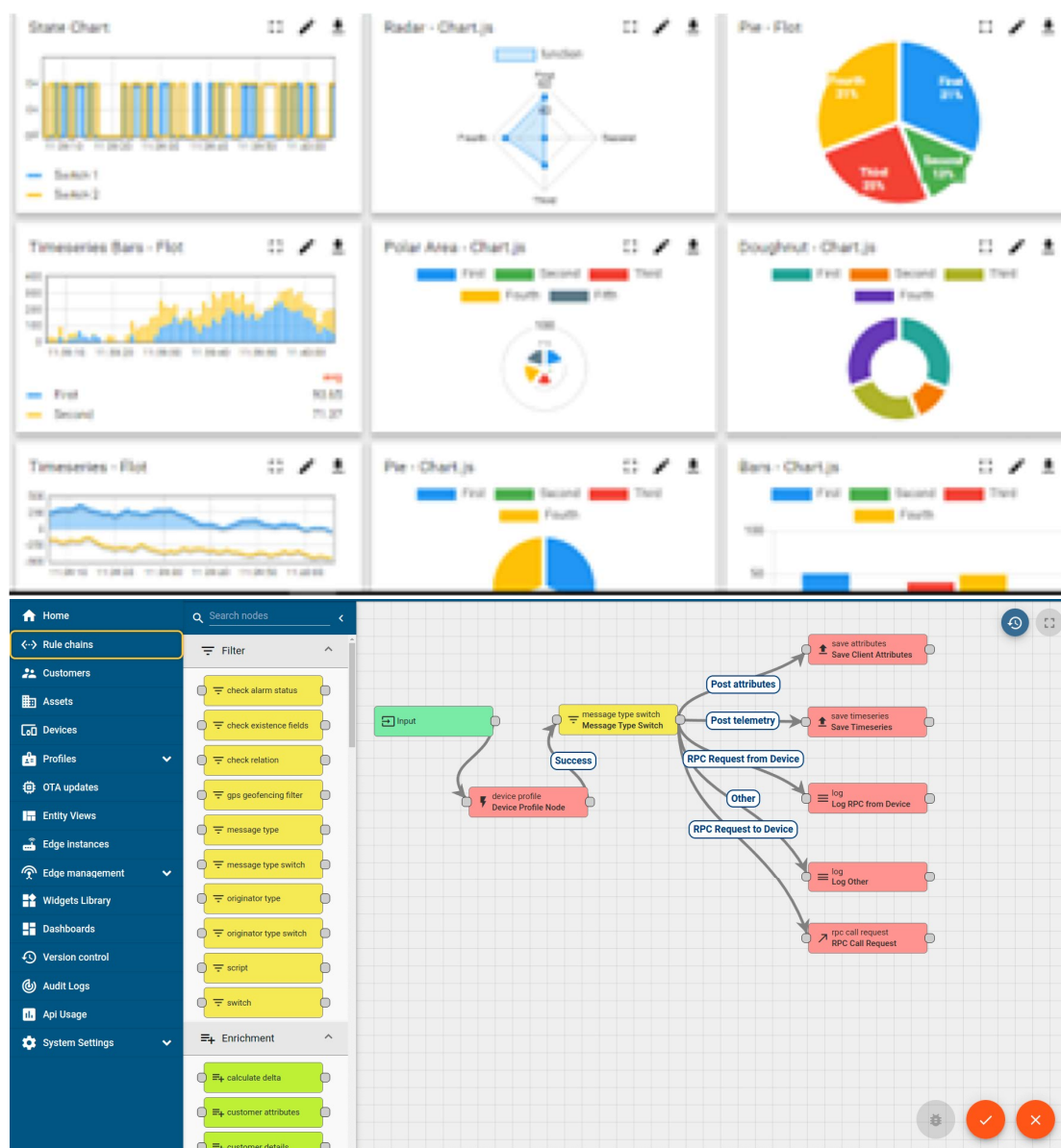
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



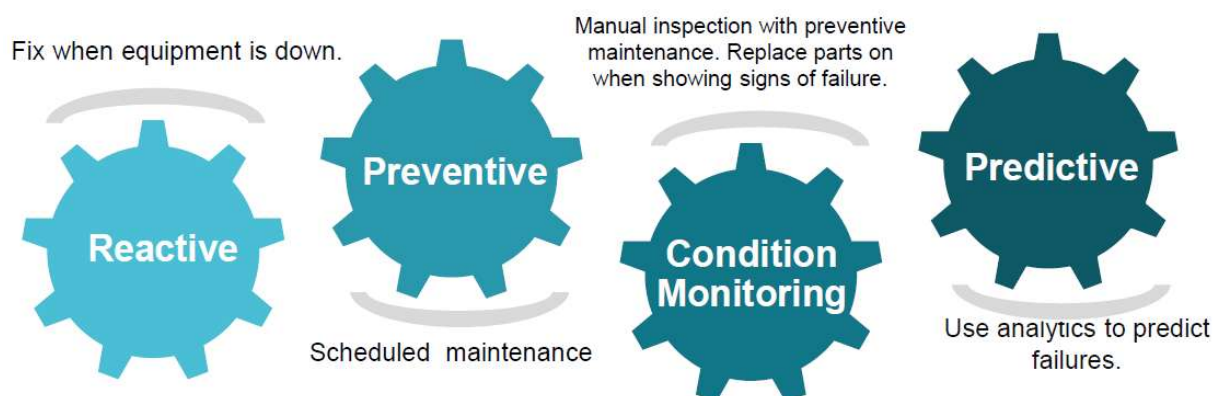


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

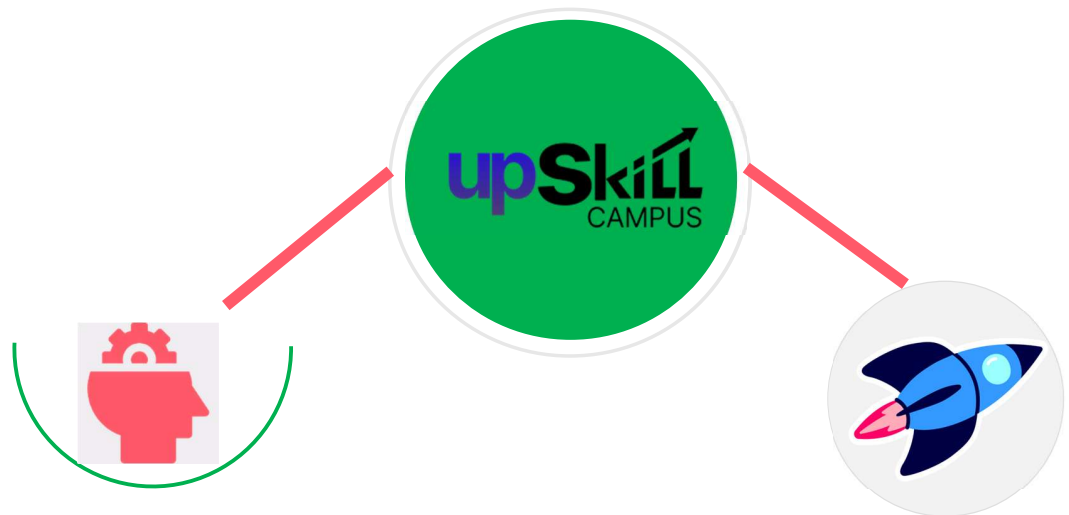
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

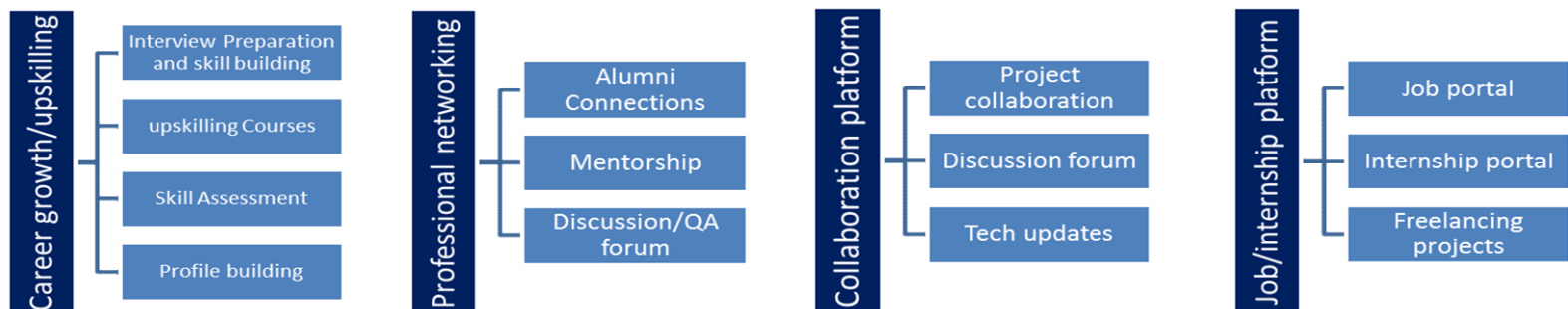
USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- To gain practical knowledge of Python Full Stack Development.
- To understand the complete software development life cycle.
- To develop a real-world web application using modern technologies.
- To strengthen programming, debugging, and problem-solving skills.
- To gain experience in database design and management.
- To improve frontend and backend integration techniques.
- To enhance teamwork, communication, and documentation skills.
- To prepare for industry-level software development practices.

2.5 Reference

- [1] Python Official Documentation – <https://docs.python.org/>
- [2] Flask Official Documentation – <https://flask.palletsprojects.com/>
- [3] SQLite Documentation – <https://sqlite.org/docs.html>
- [4] HTML5 Documentation – <https://developer.mozilla.org/>
- [5] CSS Documentation – <https://developer.mozilla.org/>
- [6] JavaScript Documentation – <https://developer.mozilla.org/>

2.6 Glossary

Terms	Acronym
Flask	Python web framework
SQLite	Lightweight relational database

CRUD	Create, Read, Update, Delete
UI	User Interface
DB	Database
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
Authentication	User identity verification
Admin Panel	Dashboard for managing quizzes and users

3 Problem Statement

In the assigned problem statement

Traditional quiz and examination systems often rely on manual processes for conducting quizzes, evaluating answers, and maintaining student records. These methods are time-consuming, prone to human error, and require significant administrative effort. Additionally, maintaining question banks and calculating scores manually becomes difficult when the number of participants increases.

To overcome these limitations, there is a need for a web-based quiz application that automates the entire quiz process. The system should provide secure user authentication, allow users to participate in quizzes online, calculate scores automatically, and enable administrators to efficiently manage quizzes, questions, and user information.

The Quiz Game project addresses these challenges by providing a user-friendly web application developed using Python Flask and SQLite. It simplifies quiz management, ensures accurate result generation, and improves the overall user experience.

4 Existing and Proposed solution

Existing Solution

Many traditional quiz systems and classroom assessments are conducted manually or through basic applications that provide limited functionality. These systems generally have the following limitations:

- Manual evaluation consumes considerable time.
- Maintaining quiz records is difficult.
- Limited security for user authentication.
- No centralized database for storing quiz information.
- Lack of automated score calculation.
- Poor user experience and limited accessibility.
- Difficulty in managing multiple quizzes and questions.

Proposed Solution

The proposed solution is a Quiz Game, a web-based application developed using Python Flask, SQLite, HTML, CSS, and JavaScript.

The application provides:

- Secure user registration and login.
- Separate Admin and User modules.
- Online quiz participation.
- Automatic score calculation.
- Result history management.
- Quiz and question management.
- Easy database maintenance.
- Responsive and user-friendly interface.

The system minimizes manual work while improving accuracy, efficiency, and accessibility.

Value Addition

Compared to existing solutions, the proposed system offers several advantages:

- Automated evaluation and instant result generation.
- Secure authentication using password hashing.
- Easy management of quizzes and questions.
- Centralized database for storing records.
- Timer-based quiz functionality.
- Improved user interface and navigation.
- Reduced administrative workload.
- Faster and more reliable quiz management.

4.1 Code submission (Github link)

<https://github.com/Dilip-S48/upskillcampus/blob/main/QuizGame.py>

4.2 Report submission (Github link)

https://github.com/Dilip-S48/upskillcampus/blob/main/QuizGame_Dilip S USC UCT.pdf

5 Proposed Design/ Model

The Quiz Game follows a client-server architecture where users interact with the application through a web browser. The frontend, developed using HTML, CSS, JavaScript, and Bootstrap, communicates with the Flask backend. Flask processes user requests, performs authentication, manages quiz data, and interacts with the SQLite database for storing and retrieving information. After processing, the server returns the appropriate response to the user.

The application consists of two major modules:

- **Admin Module**
 - Manage Users
 - Manage Quizzes
 - Manage Questions
 - View Results
- **User Module**
 - Register/Login
 - Attempt Quiz
 - View Score
 - View Previous Results

This modular architecture improves maintainability, scalability, and ease of future enhancements.

5.1 High Level Diagram

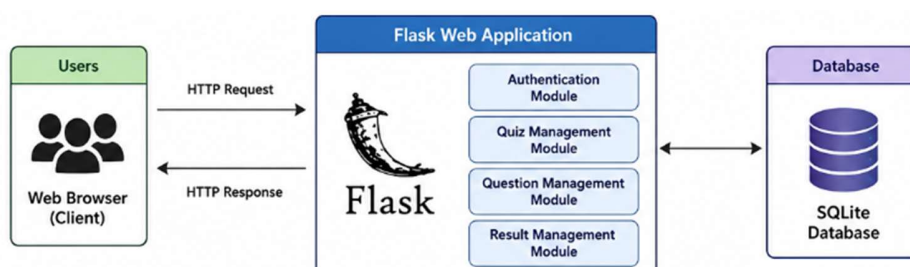


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

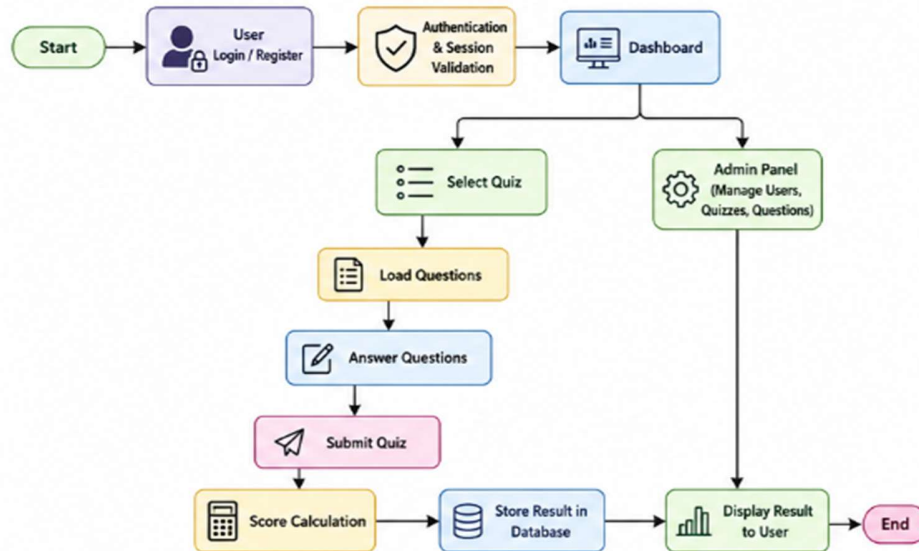


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

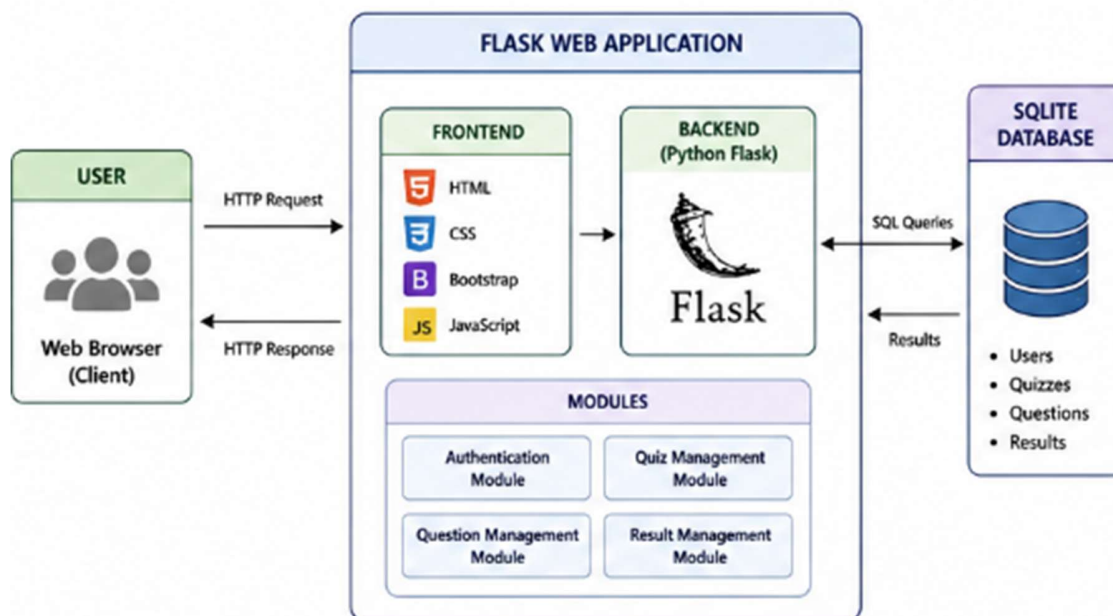


Figure 3: BLOCK DIAGRAM OF THE SYSTEM

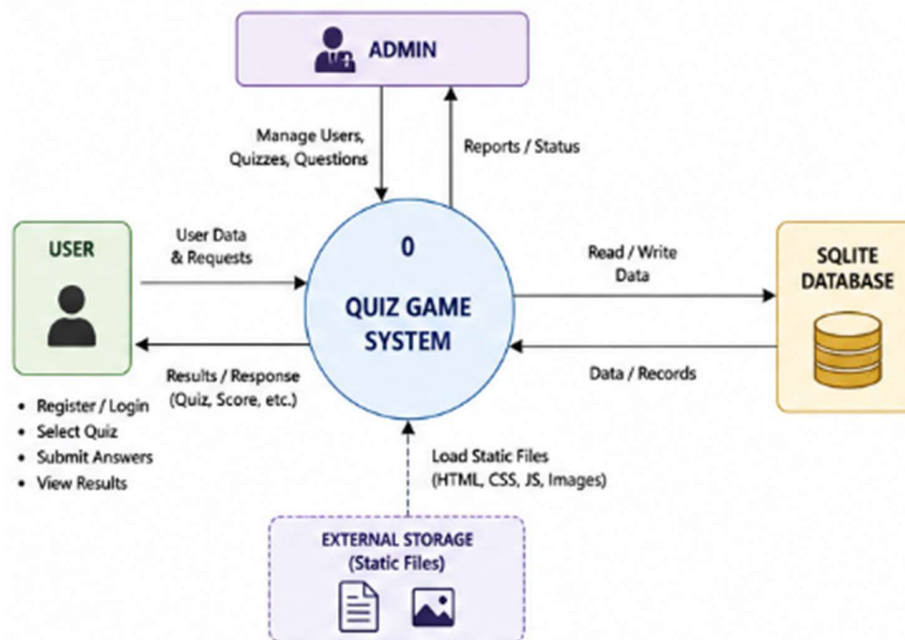


Figure 4: DATA FLOW DIAGRAM OF THE SYSTEM

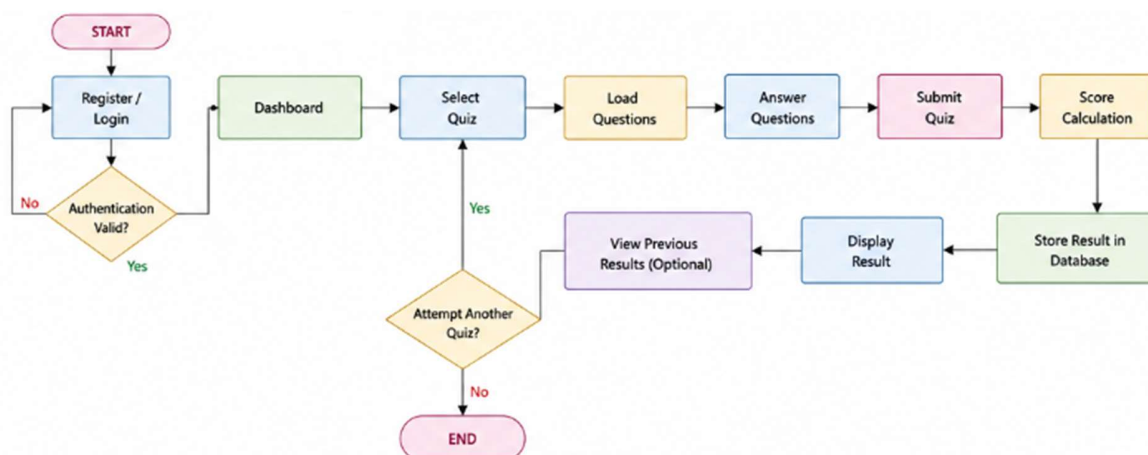
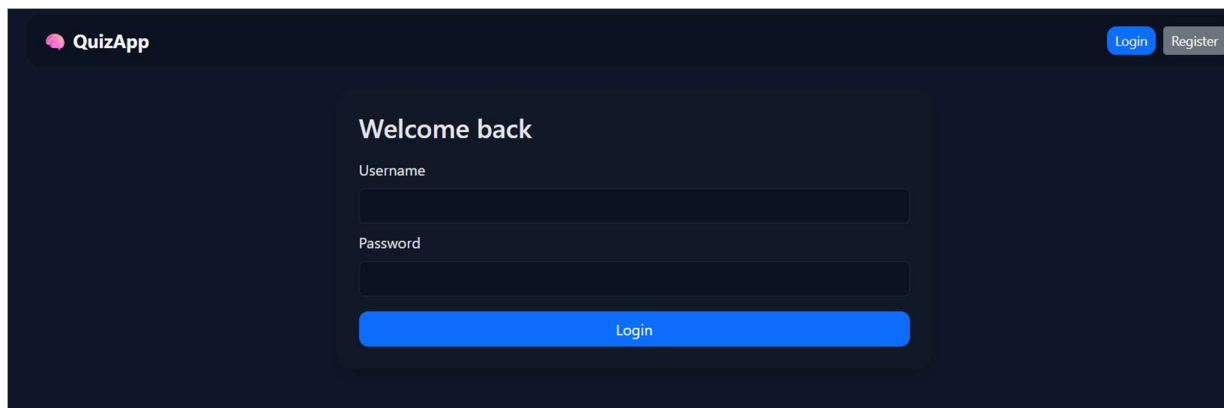


Figure 5: FLOW CHART



QuizApp Login Register

Welcome back

Username

Password

Login

Figure 6: LOGIN PAGE



QuizApp Dashboard My Results Hi, dilip Logout

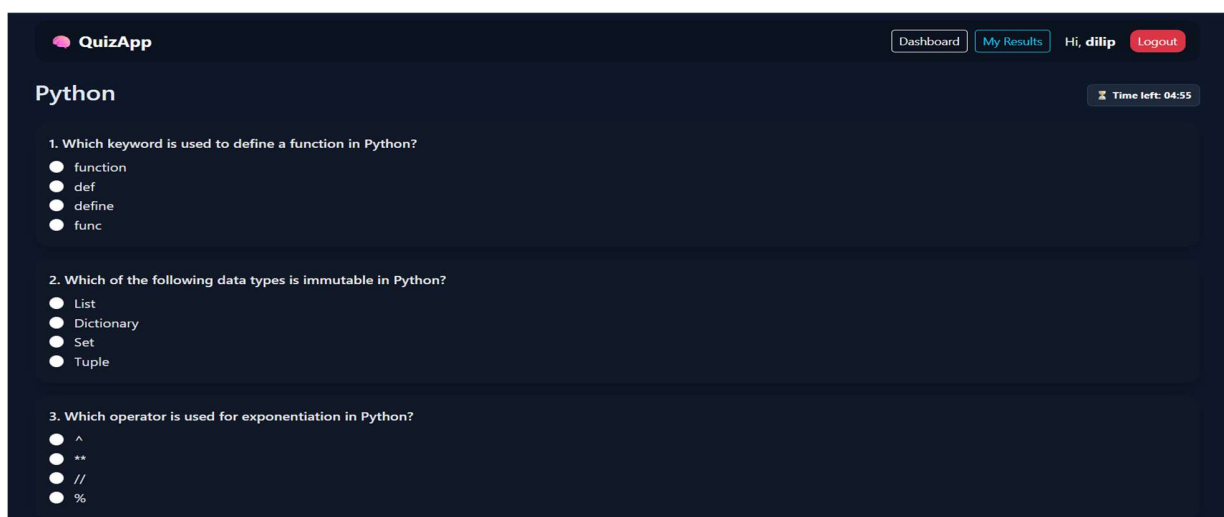
Available Quizzes

Python 5 min

5 questions

Start

Figure 7: USER DASHBOARD



QuizApp Dashboard My Results Hi, dilip Logout

Python Time left: 04:55

1. Which keyword is used to define a function in Python?

- ☐ function
- ☐ def
- ☐ define
- ☐ func

2. Which of the following data types is immutable in Python?

- ☐ List
- ☐ Dictionary
- ☐ Set
- ☐ Tuple

3. Which operator is used for exponentiation in Python?

- ☐ ^
- ☐ **
- ☐ //
- ☐ %

Figure 8: QUIZ INTERFACE

QuizApp Dashboard My Results Hi, dilip Logout

My Results

Quiz	Score	Total	Percentage	Details
Python	5	5	100.0%	Show

Figure 9: QUIZ RESULT INTERFACE

QuizApp Dashboard My Results Hi, admin Logout

Users 2 Admin

Quizzes 1

Attempts 2

Manage Users

Available Quizzes

+ Create Quiz

Python 5 min

5 questions

Start Manage Delete

Figure 10: ADMIN DASHBOARD

QuizApp Dashboard My Results Hi, admin Logout

Manage: Python

5 min

Edit Quiz Details

Quiz Title
Python

Duration (minutes)
5

Update

Add Question

Question

Option A Option B

Option C Option D

Correct Answer
A

Questions (5)

#	Question	Correct	Actions
1	Which keyword is used to define a function in Python?	B	Edit Delete
2	Which of the following data types is immutable in Python?	D	Edit Delete
3	Which operator is used for exponentiation in Python?	B	Edit

Figure 11: QUIZ MANAGEMENT INTERFACE

6 Performance Test

The performance of the Quiz Game application was evaluated to ensure that all functionalities operated correctly and efficiently. The testing focused on user authentication, quiz execution, score calculation, database operations, and overall application responsiveness. The application was tested under different user scenarios to verify its reliability and accuracy.

The major constraints considered during testing were:

- User authentication accuracy
- Quiz loading speed
- Automatic score calculation
- Database read/write operations
- User interface responsiveness
- Data consistency and reliability

The application successfully handled these constraints and produced accurate results during testing.

6.1 Test Plan/ Test Cases

Test Case ID	Test Description	Expected Result	Status
TC-01	User Registration	User account created successfully	Pass
TC-02	User Login	Valid user logs into dashboard	Pass
TC-03	Invalid Login	Error message displayed	Pass
TC-04	Load Quiz	Quiz questions displayed correctly	Pass
TC-05	Submit Quiz	Answers submitted successfully	Pass
TC-06	Score Calculation	Correct score generated automatically	Pass
TC-07	View Results	User result displayed correctly	Pass

Test Case ID	Test Description	Expected Result	Status
TC-08	Admin Login	Admin dashboard accessible	Pass
TC-09	Add Quiz	Quiz added successfully	Pass
TC-10	Delete Quiz	Quiz removed successfully	Pass

6.2 Test Procedure

The following testing procedure was followed:

1. Created multiple user accounts for testing.
2. Verified user registration and login functionality.
3. Tested quiz loading and navigation between questions.
4. Submitted quizzes with different answer combinations.
5. Verified automatic score calculation.
6. Checked whether quiz results were stored correctly in the SQLite database.
7. Tested admin functionalities such as adding, editing, and deleting quizzes and questions.
8. Repeated testing after fixing identified issues to ensure application stability.

6.3 Performance Outcome

The performance evaluation demonstrated that the Quiz Game application met the expected functional requirements.

Test Results

Parameter	Outcome
User Authentication	Successful
Quiz Loading Speed	Fast (Less than 2 seconds)

Parameter	Outcome
Score Calculation	100% Accurate
Database Operations	Successful
UI Responsiveness	Smooth
Result Generation	Instant
Overall Functionality	Successful

Performance Summary

- The application successfully authenticated users without errors.
- Quiz questions loaded quickly and displayed correctly.
- Automatic score calculation produced accurate results.
- SQLite database operations were performed efficiently.
- The user interface responded smoothly during testing.
- All major functionalities passed testing without critical failures.

Conclusion

The Quiz Game application performed reliably under normal operating conditions. The testing confirmed that the system is capable of providing secure authentication, efficient quiz management, accurate score calculation, and dependable result storage. The application meets the intended functional requirements and is suitable for educational and demonstration purposes.

7 My learnings

The Industrial Internship provided me with valuable practical experience in Python Full Stack Development and enhanced both my technical and professional skills. During the development of the Quiz Game project, I gained hands-on experience in designing, developing, testing, and deploying a web application using modern technologies.

The key learnings from this internship include:

- Strengthened my knowledge of Python programming and object-oriented programming concepts.
- Learned to develop dynamic web applications using the Flask framework.
- Gained practical experience in designing and managing databases using SQLite and Flask-SQLAlchemy.
- Improved my frontend development skills using HTML, CSS, Bootstrap, and JavaScript.
- Understood user authentication, session management, and secure password hashing techniques.
- Learned to perform CRUD (Create, Read, Update, Delete) operations efficiently.
- Improved debugging, testing, and troubleshooting skills during application development.
- Understood the importance of proper project planning, documentation, and version control.
- Enhanced analytical thinking and problem-solving abilities by implementing real-world project requirements.
- Improved communication, time management, and self-learning skills while completing the internship.

This internship significantly increased my confidence in developing full-stack web applications and provided practical knowledge that will help me in future software development roles.

8 Future work scope

Although the Quiz Game application successfully fulfils its primary objectives, several additional features can be implemented in future versions to enhance its functionality and user experience.

The future enhancements include:

- Introduce multiple quiz categories such as Python, Java, C++, Aptitude, and General Knowledge.
- Implement different difficulty levels (Easy, Medium, and Hard).
- Add a leaderboard to display top-performing users based on quiz scores.
- Generate detailed performance reports with graphical analysis of user progress.
- Allow administrators to upload questions in bulk using CSV or Excel files.
- Implement email verification and password reset functionality.
- Add profile management features for users.
- Integrate a cloud database such as MySQL or PostgreSQL for improved scalability.
- Deploy the application on cloud platforms such as Render, Railway, or AWS for online accessibility.
- Implement secure HTTPS communication and role-based authorization for enhanced security.
- Introduce AI-generated quiz questions and personalized quiz recommendations based on user performance.

These enhancements would make the application more scalable, feature-rich, and suitable for use in educational institutions, online learning platforms, and recruitment assessment systems.